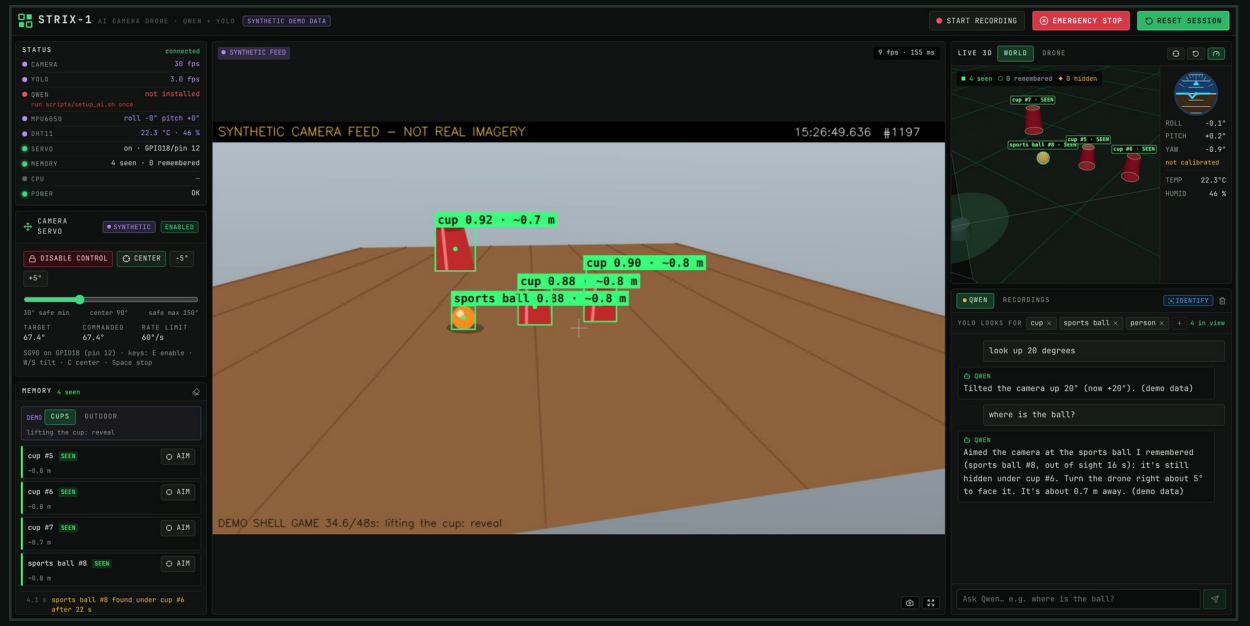
Sense. Understand. Remember. Advise.

Everything runs on the Raspberry Pi 5 on the drone: camera, sensors, object memory and a local Qwen model. No cloud.

Camera Module 3 · YOLO · MPU6050 · DHT11 · SG90 servo · Qwen



The STRIX-1 console in a browser (demo mode)

THE LOOP

One loop, all on the drone

Sensors in, AI on the Raspberry Pi 5, actions and advice out.

SENSE

Camera Module 3 Wide
live video, 102° lens

MPU6050 IMU
motion, 100 times a second

DHT11
temperature + humidity

THINK · RASPBERRY PI 5

YOLO
finds objects + distance

Sensor fusion
roll · pitch · yaw

Object memory
seen · remembered · hidden

Qwen (local)
plans, sees, answers

ACT

SG90 camera servo
aims the camera

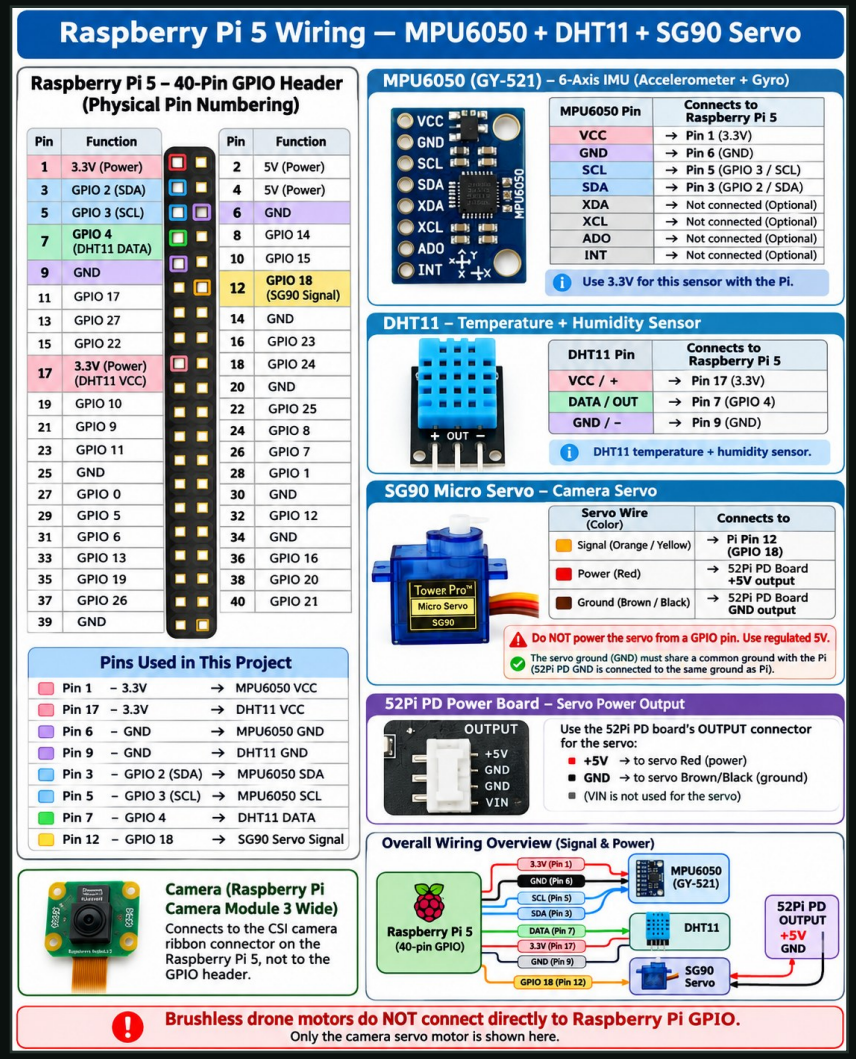
Advice for the pilot
turn, move closer, level

Live console
any browser on the Wi-Fi

IMU 100 Hz · YOLO about 3 Hz · live updates 20 Hz · the pilot always flies the drone

THE HARDWARE

What's on the drone



SG90 camera servo

Signal on GPIO18 (pin 12). Powered from the 52Pi PD board's 5 V, sharing ground with the Pi.

MPU6050 IMU

I2C: SDA pin 3, SCL pin 5, 3.3 V on pin 1, ground on pin 6.

DHT11 temperature + humidity

Data on GPIO4 (pin 7), 3.3 V on pin 17, ground on pin 9. Read by the Pi's kernel driver.

Camera Module 3 Wide

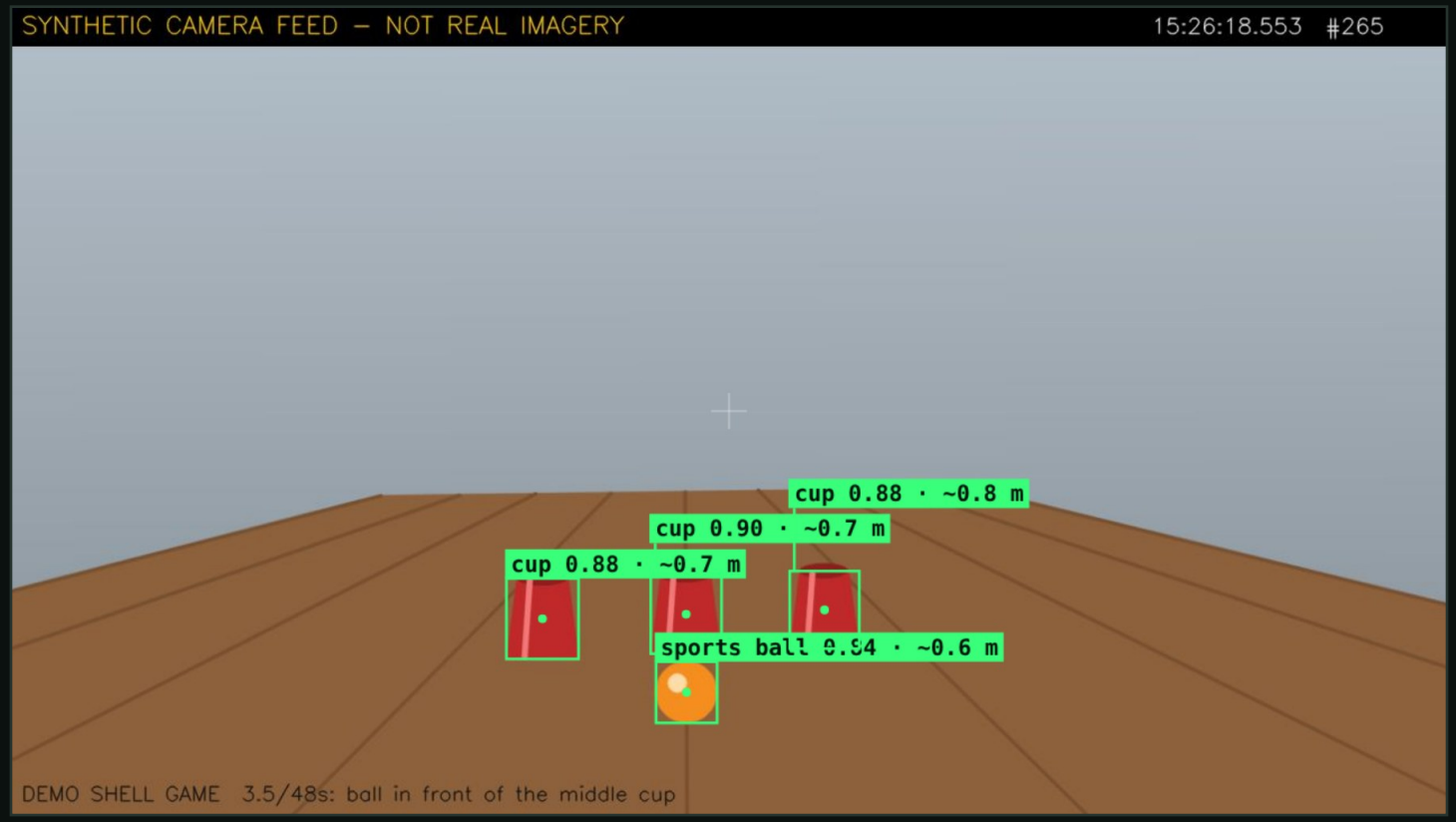
CSI ribbon cable, 102° field of view, streamed at 720p.

Raspberry Pi 5 (16 GB)

On a 250 mm ZMR250 frame, powered by a USB-C PD power bank.

IT SEES

YOLO finds objects and estimates distance



Object detection

YOLOX-Tiny runs on the Pi about 3 times a second. An optional model adds 600 everyday objects, including pens.

Distance from the lens

A known object size and the 102° lens give the distance:

```
distance ≈ f · H / h
```

Honest numbers

Every distance is shown as an estimate (about ±30 %), like “cup ~0.7 m”.

IT FEELS THE DRONE

Motion and air from real sensors



MPU6050 IMU, 100 times a second

Accelerometer and gyroscope fused into roll, pitch and yaw. The 3D drone moves with the real one.

Used everywhere

It keeps remembered objects in the right place when the drone turns, and lets Qwen say “level the drone first”.

DHT11 temperature and humidity

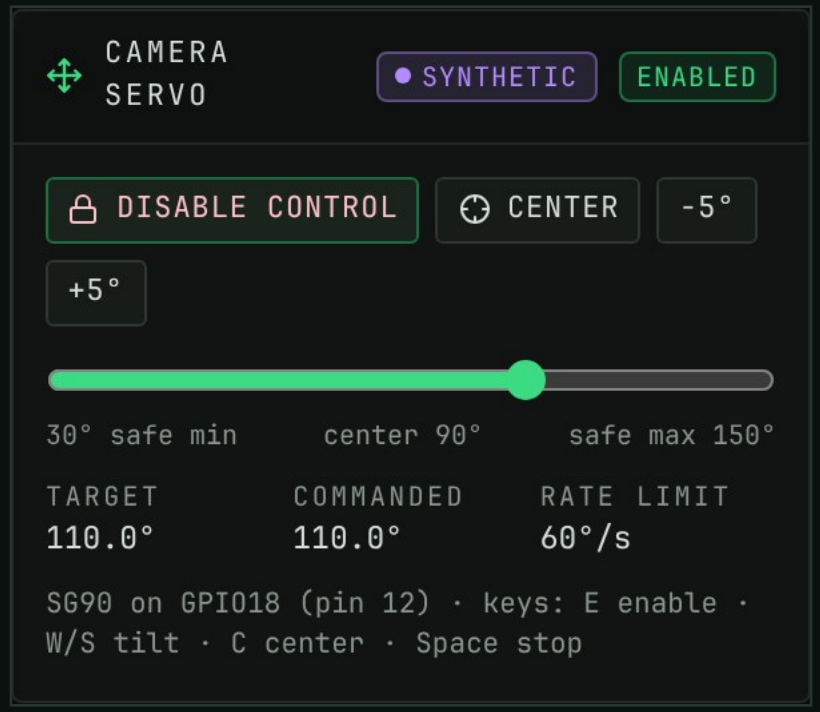
Read by the Pi's kernel driver every 3 seconds, with session min and max.

Grounded answers

Ask “how warm is it?” and Qwen answers from the sensor, not from a guess.

IT AIMS THE CAMERA

Qwen controls the SG90 servo



Safe by design: limits 30–150°, at most 60° per second, and EMERGENCY STOP (Space) turns it off until the pilot enables it again.

1 You ask

“Find the person and keep them centered.”

2 Qwen plans

```
{"plan": [{"skill": "find", "target": "person"}, {"skill": "track", "target": "person", "seconds": 8}]}
```

3 The servo moves

Tilts the camera until the person is centered, then Qwen answers with distance and direction.

IT REMEMBERS

Object permanence: seen, remembered, hidden

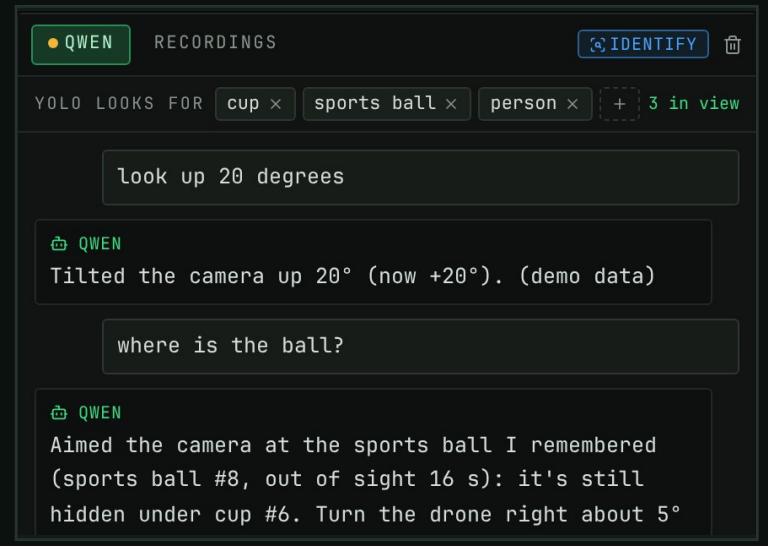
SEEN in view now | REMEMBERED out of view, with a timer | HIDDEN under a cup, moves with it



1 A cup covers the ball: it's marked HIDDEN.



2 The cups are shuffled: the ball moves with its cup.



3 “Where is the ball?” The camera aims back at the right cup.

When the cup is lifted: “found under cup after 22 s — memory was right.”

BUILT FOR THE EDGE

Small, fast, cool enough to fly

0
cloud calls: video never leaves the drone

2B
parameter Qwen vision model, local

100 Hz
IMU readings

160+
automated tests

Heat guard

The Pi watches its own temperature and slows the AI down before it overheats.



Defaults chosen for heat: 720p camera stream · light YOLO model · 2B Qwen · text-only automatic comments

WHY IT MATTERS

Impact on today's world

Search, rescue and inspection

Drones already help search for people and inspect roofs, bridges and crops. A co-pilot that remembers what it saw, even after it leaves the frame, makes that job easier.

Private and offline

No cloud: the video stays on the drone, and it keeps working where there is no signal, which is often exactly where help is needed.

Affordable and open

Built from parts anyone can buy and open-source code anyone can learn from and extend.

Trustworthy AI

It labels its estimates, never fakes GPS or battery levels, and the human pilot stays in control.

WHAT'S NEXT

Where STRIX-1 goes from here

Fly it

Validate the servo, IMU and DHT11 in flight and tune the motion filter.

Pan as well as tilt

A second servo so the camera can turn without turning the drone.

See depth

A small depth sensor for better distances and a true 3D memory.

Faster vision

A Raspberry Pi AI accelerator for quicker, cooler detection.

STRIX-1

Sees · aims · feels · remembers — on the drone.

Thank you

github.com/Nesi56/STRIX-1

Try the demo without hardware: `scripts/run_mock.sh`



Scan for the code